The function f is defined by

$$\mathrm{f}(x) = \frac{x+3}{x-4} \qquad x \in \mathbb{R}, x \neq 4$$

(a) Find $\mathrm{ff}(6)$

(2)

(b) Find f^{-1}

(3)

The function g is defined by

$$\mathrm{g}(x) = x^2 + 5 \qquad x \in \mathbb{R}, x > 0$$

(c) Find the exact value of a for which

$$\mathrm{gf}(a) = 7$$

(3)